

FMCG Multi-Country Sales Dataset Analysis

(PowerBI presentation & DAX writing)

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5.1 Purpose

As The Power BI layer was developed to transform the analytical results obtained from MySQL into an interactive business intelligence dashboard. While SQL was used for detailed analytical calculations and validation, Power BI was used for:

- Interactive exploration of sales performance
- Product and brand analysis
- Store and channel comparison
- Promotion analysis
- Inventory and stockout monitoring
- Profitability analysis
- Business-focused visualization and reporting

The Power BI report connects directly to the MySQL database through an on-premises data gateway.

5.2 PowerBI (online) Connection Scheme

The architecture I followed from MySQL to Report generation is below:

MySQL Database → On-Premises Data Gateway → Power BI Service → Semantic Model → DAX Measures → Interactive Report

Connection configuration:

Configuration	Value
Database	fmcg_recordbook
Server	localhost:3306
Source table	Sales_analysis
Gateway	FMCG_MySQL_Gateway
Connection type	MySQL
Authentication	Basic
SSL encryption	Encrypted

5.3 PowerBI Measures

I have created several DAX measures related to analytical questions to ease calculations. One thing to note that, the 23 analytical queries that I've performed in MySQL was to check the results/queries were ok or not. The actual business analysis happens in PowerBI in semantic model, DAX measures.

- Total Net Sales** = Total Net Sales = SUM (sales_analysis[net_sales])
- Total Gross Sales** = SUM (sales_analysis[gross_sales])
- Total Units Sold** = SUM (sales_analysis[units_sold])
- Total Sales Record** = COUNTROWS (sales_analysis)

- v. **Total Purchase Cost** = SUMX ('sales_analysis', sales_analysis[purchase_cost] * sales_analysis[units_sold])
- vi. **Average Selling Price** = DIVIDE ([Total Net Sales], [Total Units Sold])
- vii. **Promotional Sales** = CALCULATE ([Total Net Sales], sales_analysis[promo_flag] = 1)
- viii. **Non-promotional Sales** = CALCULATE ([Total Net Sales], sales_analysis[promo_flag] = 0)
- ix. **Promotional Sales (%)** = DIVIDE ([Promotional Sales], [Total Net Sales])
- x. **Stock-out records** = CALCULATE (COUNTROWS (sales_analysis), sales_analysis[stockout_flag] = 1)
- xi. **Stock-out rate (%)** = DIVIDE ([Stockout Rows], [Sales Rows])
- xii. **Average Stock on hand** = AVERAGE (sales_analysis[stock_on_hand])
- xiii. **Average Lead time** = AVERAGE (sales_analysis[lead_time_days])

And many more ...

5.4 Dashboard Structure

The Power BI report is organized into five business-focused pages.

Page 1 — Executive Overview: Provides a high-level view of overall business performance.

Key metrics include:

- Total Net Sales
- Total Gross Sales
- Total Units Sold
- Average Selling Price
- Stockout Rate

Key visuals include:

- Sales trend
- Sales by category
- Sales by channel
- Yearly sales performance
- Interactive filters

Page 2 — Product & Promotion: Focuses on product, category, brand and promotional performance.

Key analyses include:

- Top-selling SKUs
- Category performance
- Brand performance
- Promotional vs non-promotional sales
- Discount and sales relationships

Page 3 — Store & Channel: Examines geographic and channel-level performance.

Key analyses include:

- Top-performing stores
- City performance
- Country performance
- Channel sales & Units sold by channel

Page 4 — Inventory & Profitability: Focuses on inventory availability, stockouts, supply-chain indicators and profitability.

Key analyses include:

- Stockout rate
- Stockout events by category
- Stockouts by store
- Average inventory
- Supplier lead time
- Sales versus inventory
- Profitability metrics

Page 5 — Business Insights: Converts analytical results into management-oriented findings and suggestions.

The page focuses on:

- Sales growth opportunities
- Product concentration
- Promotional effectiveness
- Inventory availability
- Supplier performance
- Potential operational improvements

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